

Human Capital: Education and Health in Economic Development

Chapter 8

Econ 322 – The Global Economy: Trade and Development

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Based on Todaro, P. M. and Smith, C. S. (2020), *Economic Development*, 13th ed., Pearson.

8.1 Education and Health as Objectives and “Inputs”

- ▶ Health and education are important objectives of development:
 - ▶ Reflected in Amartya Sen's capabilities approach and the core values of development
 - ▶ Key individual and multidimensional indicators of development feature health and education
 - ▶ Represented in the SDGs and in the HDI
- ▶ Health and education are also important inputs in the aggregate production function



A village school in the Central African Republic.
*Wikimedia Commons, hdptcar from Bangui,
Central African Republic, CC BY-SA 2.0*

Education and Health as Joint Investments for Development

- ▶ Health and education are investments in the same individual
- ▶ Greater **health capital may improve the returns to investments in education**
 - ▶ Health is a factor in **school attendance**
 - ▶ Healthier students **learn more effectively**
 - ▶ A longer life raises the rate of return to education
 - ▶ Healthier people have lower depreciation of education capital
- ▶ Greater **education capital may improve the returns to investments in health**
 - ▶ Public health programs need knowledge learned in school
 - ▶ Basic hygiene and sanitation may be taught in school
 - ▶ Education is needed in the training of health personnel

Improving Health and Education: Why Increasing Incomes Is Not Sufficient

- ▶ Health, education, and income can function as complements
- ▶ Increases in income often do not lead to substantial increases in investment in children's education and health
- ▶ But better-educated mothers tend to have healthier children, at any income level
- ▶ Significant market failures in education and health require policy action

Income, health and education are complements: higher income alone does not guarantee more investment in children's health and schooling, so policy must act on all three.

8.2 Investing in Education and Health: The Human Capital Approach

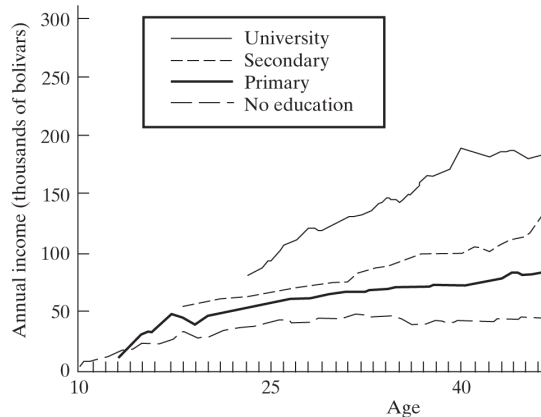
- ▶ Initial investments in health or education lead to a stream of higher future income
- ▶ The present discounted value of this stream of future income is compared to the costs of the investment
- ▶ Private returns to education are often high
- ▶ However, private returns may be higher than social returns, particularly at higher educational levels, when accounting for education subsidies



Gary Becker, pioneer of human capital theory.

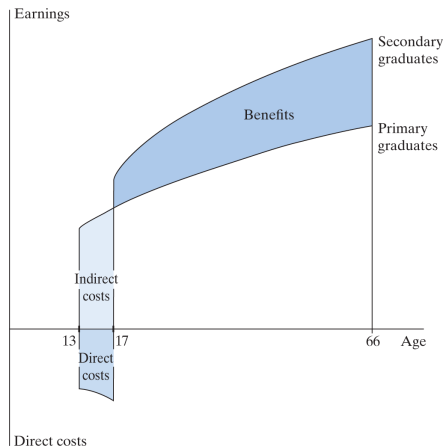
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Figure 8.1 Age-Earnings Profiles by Level of Education: Venezuela



Source: International Bank for Reconstruction and Development/The World Bank: *The Profitability of Investment in Education: Concepts & Methods* by George Psacharopoulos, 1995. Reprinted with permission.

Figure 8.2 Financial Trade-Offs in the Decision to Continue in School



Costs of staying in school now versus higher earnings later.

Source: *Todaro and Smith (2020), Figure 8.2*

Algebraic version of the schooling decision diagram

Net Present Value (NPV) of the income stream is given by:

$$NPV = \sum_t \frac{Y_t - X_t - C_t}{(1 + r)^t}$$

Where:

- ▶ t is time (year) from present. (The present is time $t = 0$.)
- ▶ Y_t is expected income from **having gone to school**, realized at time t
- ▶ X_t is expected income from having **NOT** gone to school, at time t
- ▶ C_t is the cost of going to school borne at time t (if any)
- ▶ r is the **rate of discount** used

Table 8.1 Returns to Investment in Education by Level, Regional Averages (%)

Region	Social			Private		
	Primary	Secondary	Higher	Primary	Secondary	Higher
Asia ^a	16.2	11.1	11.0	20.0	15.8	18.2
MENA ^b	15.6	9.7	9.9	13.8	13.6	18.8
Latin America/Caribbean	17.4	12.9	12.3	26.6	17.0	19.5
OECD	8.5	9.4	8.5	13.4	11.3	11.6
Sub-Saharan Africa	25.4	18.4	11.3	37.6	24.6	27.8
World	18.9	13.1	10.8	26.6	17.0	19.0

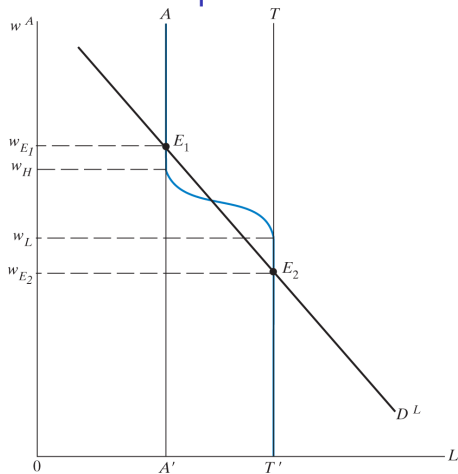
^a Non-OECD. ^b Europe/Middle East/North Africa, Non-OECD

Source: George G. Psacharopoulos and Harry A. Patrinos, "Returns to investment in education: A further update," *Education Economics* 12, No. 2 (August 2004), tab. 1. Note: How these rates of return were calculated is explained in detail in note 19 at the end of this chapter.

8.3 Child Labor: A Multiple-Equilibria Model

- ▶ **Child labor** is a widespread phenomenon
 - ▶ The problem may be modeled using a **multiple equilibria approach**
 - ▶ **Government intervention** may be called for to move to a 'better' equilibrium
 - ▶ Sometimes this shift can be self-enforcing, so active intervention is only needed at first
- ▶ **Luxury axiom:** a household with sufficiently high income would not send its children to work
 - ▶ **Substitution axiom:** adult and child labor are (perfect) substitutes; a child produces a fraction of an adult's output, $Q_C = \gamma Q_A$, $0 < \gamma < 1$
 - ▶ With sufficiently inelastic labor demand there can be multiple equilibria

Figure 8.4 Child Labor as a Bad Equilibrium



Source: From Kaushik Basu, "Child labour: Cause, consequence, and cure, with remarks on international labour standards," *Journal of Economic Literature* 37 (1999): 1101. Reprinted with the permission of the author and the American Economic Association.

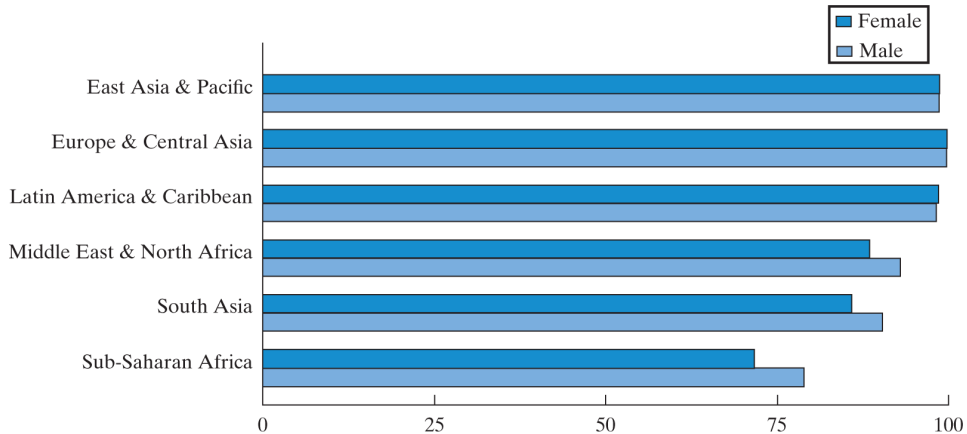
Other approaches to child labor policy

- ▶ Help **get more children into school** (as in Sustainable Development Goals), e.g., building new village schools, and providing enrollment incentives for parents
- ▶ Consider child labor an expression of poverty, so **emphasize ending poverty** generally, rather than child labor directly (a traditional World Bank approach, now modified)
- ▶ If child labor is inevitable in the short run,
regulate it to protect children and to provide support services for working children
- ▶ **Ban child labor**, or if impossible, ban child labor in its most abusive forms
- ▶ Activist approach: trade sanctions
 - ▶ Concerns: Sanctions could backfire when the result is that children shift to the informal sector, and if modern sector growth slows

The Gender Gap: Discrimination in Education and Health

- ▶ **Young females receive less education** than young males in nearly every low and lower-middle-income developing country
- ▶ Closing the educational gender gap is important in part because the social rate of return on women's education is higher than that of men:
 - ▶ If the most capable students enroll first, the marginal (next) girl enrolled is likely to be more talented than the marginal boy
 - ▶ Mothers devote a larger fraction of their income to child welfare than fathers, and education increases earnings
 - ▶ Mother's knowledge (even without extra income) helps child survival and education
 - ▶ Education for women increases productivity, thereby lowering fertility
 - ▶ Education can break the vicious cycle of poverty and inadequate schooling for girls
- ▶ Good news: Parity being approached, progress in every developing region

Figure 8.5 Youth Literacy Rate, 2016



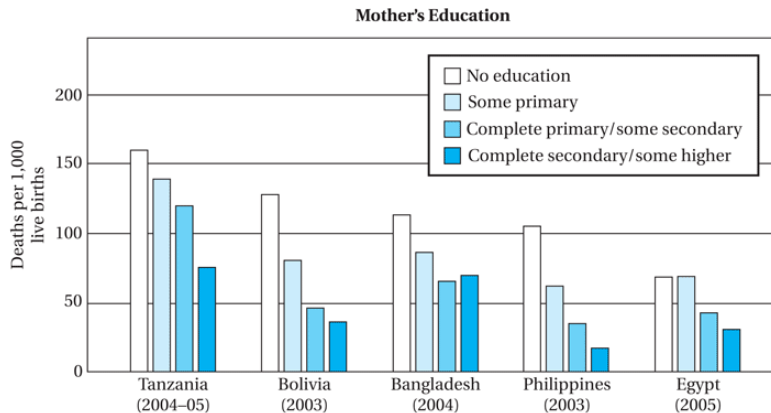
Source: UNESCO Institute for Statistics

8.5 Educational Systems and Development

- ▶ The Political Economy of Educational Supply and Demand
 - ▶ The Relationship between Employment Opportunities and Educational Demands
- ▶ Social versus Private Benefits and Costs (Figure 8.6 in the text)
 - ▶ Among factors that could augment net social returns are literacy spillover benefits
 - ▶ But net social returns are overstated by public subsidies for private education, to the extent they represent transfers for private benefits (generally larger divergences at higher educational levels)

The demand for schooling is largely a **derived demand**: families invest in education because it is the ticket to modern-sector jobs, so employment prospects drive the demand for education.

Correlation between Under-5 Mortality and Mother's Education

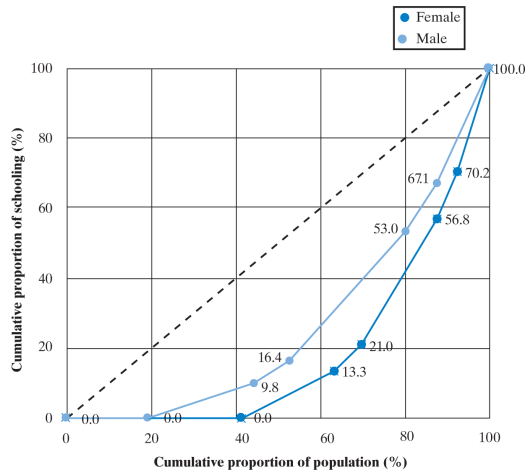


Source: International Bank for Reconstruction and Development/World Bank, *World Development Indicators*, 2007 (Washington, D.C.: World Bank, 2007), p. 119. Reprinted with permission.

Under-5 mortality falls with the mother's schooling in every country shown.

Source: *Todaro and Smith (2020)*

Figure 8.7 Lorenz Curves for Education in India

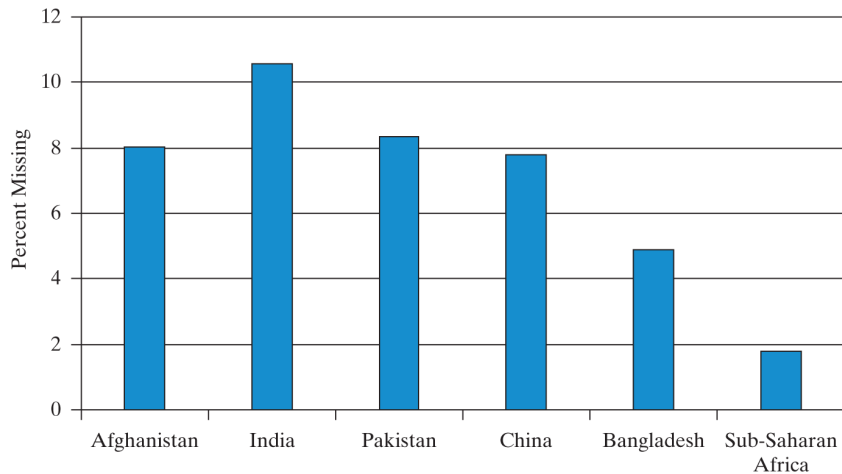


Source: <https://www.epdc.org/epdc-data-points/epdc-spotlight-india>; Data are based on the 2006 DHS India household dataset.

The “Missing Women” Crisis

- ▶ Research has concluded that **in Asia at least 100 million women are “missing”...**
 - ▶ Update: Bongaarts and Guilmoto 2015: 126 million missing globally; numbers still rising, projected to reach 150 million by 2035
- ▶ ...If gender ratios were closer to normal levels based on biology, comparing regions such as Europe, North America, and Latin America
 - ▶ That is the minimum number of additional women who would be alive in Asia
- ▶ **Some women are also missing in Africa**, but a much smaller proportion
- ▶ Reasons include inferior medical care for girls (postnatal factors), gender selective abortion (prenatal selection), or even female infanticide

Figure 8.6 Estimated Percentage of Women “Missing”



Source: United Nations, Department of Economic and Social Affairs, Population Division (2017)

8.6 Health Measurement and the Disease Burden

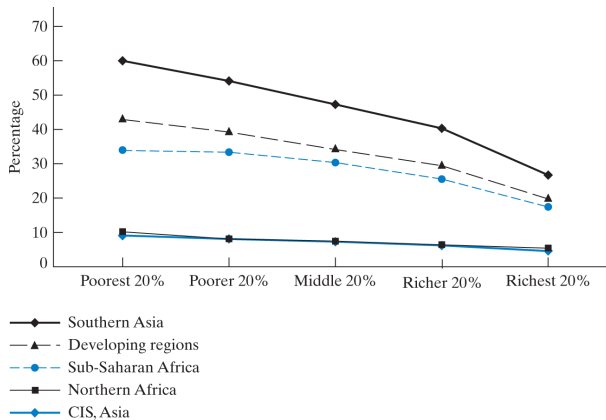
- ▶ World Health Organization (WHO): the key United Nations agency concerned with global health matters
- ▶ Disease burden in developing countries – outline:
 - ▶ HIV/AIDS
 - ▶ Malaria
 - ▶ Parasitic worms and other “neglected tropical diseases”



A child under an insecticide-treated bed net, Mwanza, Tanzania (2016).

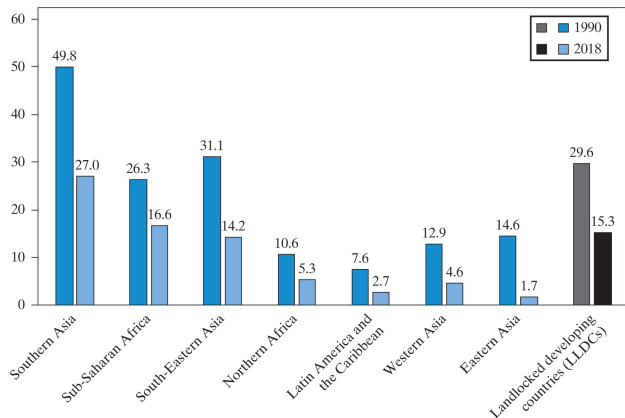
Wikimedia Commons, President's Malaria Initiative, Public domain

Figure 8.9 Underweight Children Under Five, by Household Wealth (c. 2008)



Source: From Millennium Development Goals Report, 2010, p. 14. Reprinted with permission from the United Nations.

Figure 8.10 Underweight Children Under Five, 1990 and 2018



Source: World Health Organization, 2018

Table 8.2 Some Major Neglected Tropical Diseases (1/2)

**Note:* Population considered at risk generally much higher than current prevalence; estimated DALYs are far higher than death rates

Neglected Disease	Symptoms and Effects	Global prev. (mil-lions)*	Regions with highest Prevalence/Risk
Roundworm (Ascariasis)	Malnutrition and intestinal obstruction in young children; child stunting; impaired cognition	820	East Asia and Pacific Islands, sub-Saharan Africa, India, South Asia, China, Latin America and Caribbean
Whipworm (Trichuriasis)	Colitis and inflammatory bowel disease; child stunting and impaired cognition	465	Sub-Saharan Africa, East Asia and Pacific Islands, Latin America and Caribbean, India, South Asia
Hookworm infection	Severe iron deficiency anaemia and protein malnutrition; anaemia; child stunting and impaired intellectual and cognitive development; maternal morbidity and mortality in pregnancy	439	Sub-Saharan Africa, East Asia and Pacific Islands, India, South Asia, Latin America and Caribbean
Schistosomiasis	Bladder damage, intestine or liver inflammation; chronic pain, anaemia, malnutrition and stunting; liver and intestinal fibrosis, kidney disease, female genital schistosomiasis	200+	Sub-Saharan Africa, Latin America and Caribbean

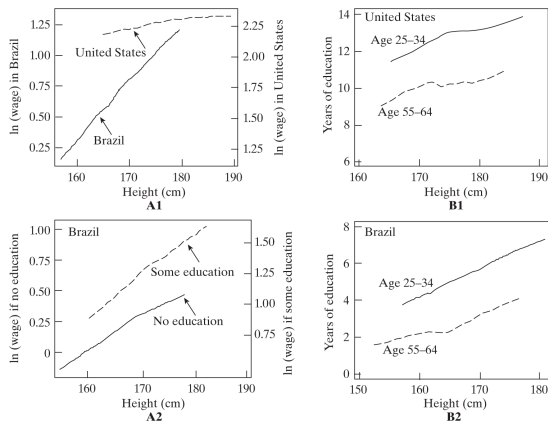
Table 8.2 Some Major Neglected Tropical Diseases (2/2)

Neglected Disease	Symptoms and Effects	Global prev. (mil-lions)*	Regions with highest Prevalence/Risk
Lymphatic filariasis (elephantiasis)	Leg swelling, disfigurement, extreme pain	120	India, South Asia, East Asia and Pacific Islands, sub-Saharan Africa
Trachoma	Blindness	60–80	Sub-Saharan Africa, Middle East and North Africa
Onchocerciasis	Larvae in skin and eyes; onchocerca skin disease; blindness	30–40	Sub-Saharan Africa, Latin America and Caribbean
Leishmaniasis	Fever, weight loss, enlargement of the spleen and liver, and anaemia	12	India, South Asia, sub-Saharan Africa, Latin America and Caribbean
Chagas' disease	Heart and digestive problems	8+	Latin America and Caribbean
Trypanosomiasis (Human African)	Sleepiness, swollen lymph nodes, weakness, psychiatric disorders, seizures	0.3	Sub-Saharan Africa

Source: World Health Organization Website, Neglected Tropical Diseases website accessed Feb. 15, 2014:

http://www.who.int/neglected_diseases/en/; infections in 2010 Rachel L Pullan, Jennifer L Smith, Rashmi Jasrasaria, and Simon J Brooke, "Global numbers of infection and disease burden of soil transmitted helminth infections in 2010," *Parasites and Vectors*, 2014; Peter Hotez, "A Plan to Defeat Neglected Tropical Diseases," *Scientific American*, Jan. 10, 2010, p90-96; Peter J Hotez, Alan Fenwick, Lorenzo Savioli, and David H Molyneux, "Rescuing the bottom billion through control of neglected tropical diseases, May 2, 2009 *Lancet* #379, p1570-75 ; Peter Hotez, "NTDs V.2.0: "Blue Marble Health"—Neglected Tropical Disease Control and Elimination in a Shifting Health Policy Landscape Blue Marble Health 2013 *PLOS Neglected Tropical Diseases*, www.plosntds.org, 1 November 2013, Vol. 7, No. 11 ,e2570; Peter Hotez et al., "Control of neglected tropical diseases," *New England Journal of Medicine*, 357: 1018–1027 (2007)

Figure 8.11 Wages, Education, and Height of Males in Brazil and the United States



Source: "Health, nutrition, and economic development," by John Strauss and Duncan Thomas, *Journal of Economic Literature* 36 (1998): 766–817. Reprinted with permission. Note: $\ln(\text{wage})$ stands for natural log of wage.

Chapter summary

- ▶ Health and education are both objectives of development and inputs into production; investments in one raise the returns to the other
- ▶ The human capital approach compares the present value of higher future earnings with the costs of schooling; private returns often exceed social returns at higher levels
- ▶ Child labor can be a bad equilibrium (luxury and substitution axioms); policy can shift the economy to a better one
- ▶ Girls receive less education, yet the social return on women's education is higher; "missing women" reflect discrimination in health
- ▶ Undernutrition and neglected tropical diseases fall hardest on the poor, and health affects wages and productivity

Concepts for Review

- ▶ Acquired immunodeficiency syndrome (AIDS)
- ▶ Basic education
- ▶ Brain drain
- ▶ Conditional cash transfer (CCT) programs
- ▶ Derived demand
- ▶ Discount rate
- ▶ Educational certification
- ▶ Educational gender gap
- ▶ Health system
- ▶ Human capital
- ▶ Human immunodeficiency virus (HIV)
- ▶ Literacy
- ▶ Neglected tropical diseases
- ▶ Private benefits of education
- ▶ Private costs of education
- ▶ Social benefits of education
- ▶ Social costs of education
- ▶ World Health Organization (WHO)